

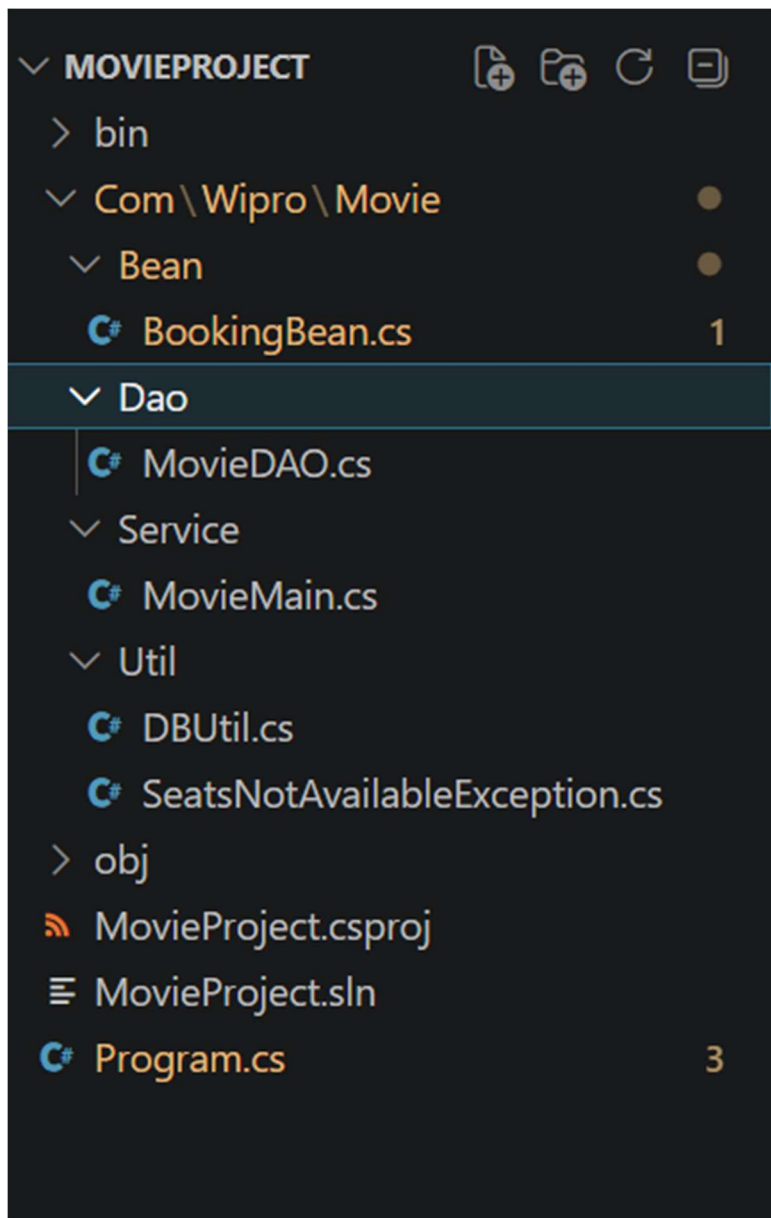
MOVIE TICKET BOOKING SYSTEM

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CST

PROJECT STRUCTURE:



MOVIE PROJECT:

Namespace: Com.Wipro.Movie.Util

Class: DBUtil

```
using Microsoft.Data.SqlClient;
namespace Com.Wipro.Movie.Util
{
    public class DBUtil
    {
        public static SqlConnection GetDBConnection()
        {
```

```

        SqlConnection con = new SqlConnection(
            "Data Source=localhost\\SQLEXPRESS;Initial Catalog=MovieDB;Integrated
Security=True;TrustServerCertificate=True");
        return con;
    }
}
}

```

Class: SeatsNotAvailableException

```

using System;
namespace Com.Wipro.Movie.Util
{
    public class SeatsNotAvailableException : Exception
    {
        public override string ToString()
        {
            return "SEATS NOT AVAILABLE";
        }
    }
}

```

Namespace: Com.Wipro.Movie.Bean

Class: BookingBean

```

using System;

namespace Com.Wipro.Movie.Bean
{
    public class BookingBean
    {
        public int BookingID { get; set; }
        public string MovieID { get; set; }
        public int TicketsBooked { get; set; }
        public DateTime BookingDate { get; set; }
    }
}

```

Namespace: Com.Wipro.Movie.Dao

Class: MovieDAO

```

using System;
using Microsoft.Data.SqlClient;
using Com.Wipro.Movie.Util;
using Com.Wipro.Movie.Bean;

namespace Com.Wipro.Movie.Dao

```

```

{
    public class MovieDAO
    {
        public int generateSequenceNumber()
        {
            Random r = new Random();
            return r.Next(1000, 9999);
        }

        public bool validateMovie(string movieID)
        {
            SqlConnection con = DBUtil.GetDBConnection();
            con.Open();

            SqlCommand cmd = new SqlCommand(
                "SELECT COUNT(*) FROM MOVIE_TBL WHERE Movie_ID=@id", con);
            cmd.Parameters.AddWithValue("@id", movieID);

            int count = (int)cmd.ExecuteScalar();
            con.Close();

            return count > 0;
        }

        public int getAvailableSeats(string movieID)
        {
            SqlConnection con = DBUtil.GetDBConnection();
            con.Open();

            SqlCommand cmd = new SqlCommand(
                "SELECT Seats_Available FROM MOVIE_TBL WHERE Movie_ID=@id", con);
            cmd.Parameters.AddWithValue("@id", movieID);

            int seats = (int)cmd.ExecuteScalar();
            con.Close();

            return seats;
        }

        public bool updateSeats(string movieID, int newSeats)
        {
            SqlConnection con = DBUtil.GetDBConnection();
            con.Open();

            SqlCommand cmd = new SqlCommand(
                "UPDATE MOVIE_TBL SET Seats_Available=@s WHERE Movie_ID=@id",
con);
            cmd.Parameters.AddWithValue("@s", newSeats);
            cmd.Parameters.AddWithValue("@id", movieID);

```

```

        int rows = cmd.ExecuteNonQuery();
        con.Close();

        return rows > 0;
    }

    public bool bookTicket(BookingBean bookingBean)
    {
        SqlConnection con = DBUtil.GetDBConnection();
        con.Open();

        SqlCommand cmd = new SqlCommand(
            "INSERT INTO BOOKING_TBL VALUES (@bid,@mid,@tickets,@date)", con);

        cmd.Parameters.AddWithValue("@bid", generateSequenceNumber());
        cmd.Parameters.AddWithValue("@mid", bookingBean.MovieID);
        cmd.Parameters.AddWithValue("@tickets", bookingBean.TicketsBooked);
        cmd.Parameters.AddWithValue("@date", DateTime.Now);

        int rows = cmd.ExecuteNonQuery();
        con.Close();

        return rows > 0;
    }
}
}

```

Namespace: Com.Wipro.Movie.Service

Class: MovieMain

```

using System;
using Com.Wipro.Movie.Dao;
using Com.Wipro.Movie.Bean;
using Com.Wipro.Movie.Util;

namespace Com.Wipro.Movie.Service
{
    public class MovieMain
    {
        MovieDAO dao = new MovieDAO();

        public string viewMovie(string movieID)
        {
            if (dao.validateMovie(movieID))
                return "MOVIE DETAILS FOUND";
            else
                return "INVALID MOVIE ID";
        }
    }
}

```

```

public string book(BookingBean bookingBean)
{
    if (bookingBean == null)
        return "INVALID";

    if (!dao.validateMovie(bookingBean.MovieID))
        return "INVALID MOVIE ID";

    try
    {
        int seats = dao.getAvailableSeats(bookingBean.MovieID);

        if (seats < bookingBean.TicketsBooked)
            throw new SeatsNotAvailableException();

        int newSeats = seats - bookingBean.TicketsBooked;
        dao.updateSeats(bookingBean.MovieID, newSeats);

        if (dao.bookTicket(bookingBean))
            return "SUCCESS";
    }
    catch (SeatsNotAvailableException e)
    {
        return e.ToString();
    }

    return "FAILED";
}
}

```

Main Method (Program.cs):

```

using System;
using Com.Wipro.Movie.Service;
using Com.Wipro.Movie.Bean;
class Program
{
    static void Main(string[] args)
    {
        MovieMain movieMain = new MovieMain();

        while (true)
        {
            Console.WriteLine("\n MOVIE TICKET SYSTEM ");
            Console.WriteLine("1. View Movie");
            Console.WriteLine("2. Book Ticket");
            Console.WriteLine("3. Exit");
            Console.Write("Enter choice: ");

```

```

int choice = Convert.ToInt32(Console.ReadLine());

switch (choice)
{
    case 1:
        Console.Write("Enter Movie ID: ");
        string movieId = Console.ReadLine();

        Console.WriteLine(movieMain.viewMovie(movieId));
        break;

    case 2:
        BookingBean b = new BookingBean();

        Console.Write("Enter Movie ID: ");
        b.MovieID = Console.ReadLine();

        Console.Write("Enter number of tickets: ");
        b.TicketsBooked = Convert.ToInt32(Console.ReadLine());

        b.BookingDate = DateTime.Now;

        Console.WriteLine(movieMain.book(b));
        break;

    case 3:
        return;

    default:
        Console.WriteLine("Invalid choice");
        break;
}
}
}
}

```

Table Name: MOVIE_TBL

	Movie_ID	Movie_Name	Language	Seats_Available
1	M101	Leo	Tamil	50
2	M102	Oppenheimer	English	0

OUTPUT:

Test Case:1

```
PS C:\Users\vidhya shree\Downloads\MovieProject\MovieProject> dotnet run

--- MOVIE TICKET SYSTEM ---
1. View Movie
2. Book Ticket
3. Exit
Enter choice: 1
Enter Movie ID: M101
MOVIE DETAILS FOUND
```

Test Case:2

```
--- MOVIE TICKET SYSTEM ---
1. View Movie
2. Book Ticket
3. Exit
Enter choice: 2
Enter Movie ID: M101
Enter number of tickets: 2
SUCCESS
```

Test Case:3

```
--- MOVIE TICKET SYSTEM ---
1. View Movie
2. Book Ticket
3. Exit
Enter choice: 2
Enter Movie ID: M102
Enter number of tickets: 4
SEATS NOT AVAILABLE
```


Test Case:4

```
--- MOVIE TICKET SYSTEM ---  
1. View Movie  
2. Book Ticket  
3. Exit  
Enter choice: 1  
Enter Movie ID: M104  
INVALID MOVIE ID
```